

Comparative Study Between MALDI-TOF-MS and HPLC-DAD-TOF-MS Techniques in Identifying Dyes in A Heritage Caucasian Carpet

Comparative Study Between MALDI-TOF-MS and HPLC-DAD-TOF-MS Techniques in Identifying Dyes in A Heritage Caucasian Carpet

دراسة مقارنة بين تقنيتي MALDI-TOF-MS وتقنية HPLC-DAD-TOF-MS في التعرف على الصبغات في إحدى السجاجيد القوقازية التراثية

Nabil Mabrouk

Faculty of Archaeology - Damietta University

nsh00@du.edu.eg

Abstract

Variant techniques have been utilized in the identification of dyes in heritage textiles and carpets. Mass spectrometry is the most used one either in an invasive way by extraction of dyes from fibers or in a non-invasive way by direct analysis without extraction. The present work aims at presenting a pioneer comparative study between both techniques in identifying dyes in a heritage Caucasian carpet preserved in the Agricultural Museum in Egypt. The study used MALDI-TOF-MS as a non-invasive technique and HPLC-DAD-TOF-MS as an invasive one to clarify the points of strength and weakness in each method, and to recommend which one we may use, and in which object. The obtained results prioritized using MALDI-TOF-MS due to its easiness, fastness, cost-effectiveness, no chemicals or hazards, and almost no sampling. HPLC-DAD-TOF-MS proved that it is more accurate and universal and able to detect more components. MALDI-TOF-MS was able to efficiently identify madder and indigo/woad in almost all samples, so it is recommended in these cases. On the contrary, HPLC-DAD-TOF-MS is recommended in case of MALDI-TOF-MS failure, or in distinguishing between dyes, especially weld, fustic, and other dyes, which were efficiently characterized only by HPLC-DAD-TOF-MS.

Keywords: Mass spectrometry, MALDI-TOF-MS, HPLC-DAD-TOF-MS, Conservation, Heritage carpet.

الملخص

تستخدم العديد من التقنيات في التعرف على الصبغات في المنسوجات والسجاجيد التراثية. يُعتبر مطياف الكتلة هو الأكثر استخدامًا إما بطريقة متلفة عن طريق استخلاص الصبغات من الألياف أو بطريقة غير متلفة عن طريق التحليل المباشر دون استخلاص يهدف هذا لعملي إلى تقديم دراسة مقارنة رائدة بين التقنيتين في تحديد الصبغات المستخدمة في سجادة قوقازية تراثية محفوظة في المتحف الزراعي في مصر. استخدمت الدراسة تقنيتي MALDI-TOF-MS كطريقة غير متلفة وتقنية HPLC-DAD-TOF-MS كطريقة متلفة لتوضيح نقاط القوة والضعف في كلتا الطريقتين، ولتوصية أيهما يمكن استخدامه، وفي أي أثر. أعطت النتائج التي تم الحصول عليها الأولوية لاستخدام MALDI-TOF-MS بسبب سهولته وسرعته وفعاليته وكلفته المنخفضة وخلو الطريقة من الكيماويات والمخاطر وعدم أخذ عينات تقريباً. أثبتت تقنية HPLC-DAD-TOF-MS أنها الأكثر دقة وتنوعاً وقدرة على كشف المزيد من مكونات الصبغات. تمكنت تقنية MALDI-TOF-MS من التعرف بكفاءة على القوة والنيلة/الوسمة في جميع العينات تقريباً، لذا يُوصى بها في هذه الحالات. على عكس ذلك، يُوصى باستخدام HPLC-DAD-TOF-MS حال فشل MALDI-TOF-MS أو في التمييز بين بعض الصبغات مثل صبغات البليحة، والفسطيط وغيرها من الصبغات التي تم تحليلها بكفاءة بواسطة HPLC-DAD-TOF-MS.

الكلمات الدالة: مطياف الكتلة، MALDI-TOF-MS، HPLC-DAD-TOF-MS، صيانة، السجاد الأثري.

1. Introduction

The first step in the conservation of heritage textiles is to identify their components. This step plays an important role in preserving the properties and appearance of the heritage textiles, consequently, helping the conservators to implement high-quality conservation works, and revealing more secrets of the ancient civilizations¹. Our ancestors used more than 50 species of vegetable dye, and others of animal sources, and different dying techniques and mordants to make different shadows and colors. Therefore, techniques used in identifying these dyes used in heritage textiles should enable the identification of different molecules in different dyes, regardless of the dyed fibers^{2,3,4}. Due to the different nature of different molecules, many analytical techniques can be used. Most of the heritage materials are preferred to be analyzed with non-invasive and micro-invasive analytical techniques⁵.

The earliest analytical techniques used in the identification of dyes in heritage textiles included extractions and chemical reactions, visual color matching, UV-Vis spectroscopy, Paper Chromatography, and Fourier Transform Infrared (FTIR) spectroscopy^{6,7}. Other techniques such as Nuclear magnetic resonance (NMR), Mass Spectroscopy techniques (SIMS, ICP-MS, etc.), Chromatographic techniques (HPLC, TLC, GC), Inductively Coupled Plasma Optical Emission spectroscopy (ICP-OES), etc., were used^{8,9,10}. Despite most of these techniques are described as invasive/destructive/micro-destructive methods, as they require micro-samples to be collected from the heritage textile, they may be conducted using a small sample in a short time, resulting in very precise data, which helps the identification of dying molecules even if they are deteriorated by natural aging procedures^{11,12,13}. It is well-

¹ Pour, R.A., Gashti, M.P. and He, J., State-of-the-Art Characterization Methods for Historic Textiles. *Handbook of Museum Textiles*, (2022) 2, pp.107-129.

² Karapanagiotis, et al., Identification of dyes in Coptic textiles from the museum of faculty of archaeology, Cairo university. *Heritage*, (2021), 4(4), pp.3147-3156.

³ Kramell, A.E., et al., Fast direct detection of natural dyes in historic and prehistoric textiles by flowprobe™-ESI-HRMS. *RSC Advances*, (2017), 7(21), pp.12990-12997.

⁴ Kim, J.P. and Lee, J.J., *Natural dyes in Korea*, Seoul National University Press, Seoul, Korea (2003), pp. 25-28.

⁵ De Ferri, L. and Andriulo, F., Non-Invasive Analytical Techniques for the Study of Dyes and Pigments in Historical Textiles. *Handbook of Museum Textiles*, 2, (2022), pp.171-190.

⁶ Tiedemann, E.J. and Yang, Y., Fiber-safe extraction of red mordant dyes from hair fibers. *Journal of the American Institute for Conservation*, (1995), 34(3), pp.195-206.

⁷ Gillard, R.D., et al., The detection of dyes by FTIR microscopy. *Studies in conservation*, (1994), 39(3), pp.187-192.

⁸ De Ferri, L. and Andriulo, F., Non-Invasive Analytical Techniques for the Study of Dyes and Pigments, pp.171-190.

⁹ Poulin, J., A new methodology for the characterisation of natural dyes on museum objects using gas chromatography–mass spectrometry. *Studies in Conservation*, (2018), 63(1), pp.36-61.

¹⁰ Nedumpillil, N.N., et al., Destructive analytical techniques for the analysis of historic textiles. *Handbook of Museum Textiles*, (2022), 2, pp.215-234.

¹¹ Poulin, J., A new methodology for the characterisation of natural dyes, pp.36-61.

Comparative Study Between MALDI-TOF-MS and HPLC-DAD-TOF-MS Techniques in Identifying Dyes in A Heritage Caucasian Carpet

known that the destructive/micro-destructive methods are more informative techniques¹⁴. Researches mainly found that alizarin and purpurin are the main compounds that have been identified in heritage textiles dyed with madder for red shadows, in addition to other few compounds^{15,16}. Cochineal and other insect dyes have always been used in dyeing heritage textiles in red colors¹⁷. Natural indigo and woad were used as blue dyes^{18,19,20}. Weld, Persian berries, turmeric, and saffron were detected as the prevalent dyes used in yellow shadows^{21,22,23}.

During the past few decades, researches have focused on non-invasive (maybe described as non-destructive) techniques through the application of some spectrometric methods. In all applied techniques, the identification of dye source usually depends on reference substances or library. This process isn't so easy in case of less dye compounds, or material degradation by ageing^{24,25}. Thus, the identification of dyes in

¹²De Ferri, L. and Andriulo, F., Non-Invasive Analytical Techniques for the Study of Dyes and Pigments, pp.171-190.

¹³ Nedumpillil, N.N., et al., Destructive analytical techniques for the analysis of historic textiles, pp.215-234.

¹⁴ Shahid, M., et al., Analytical methods for determination of anthraquinone dyes in historical textiles: A review. *Analytica Chimica Acta*, (2019), 1083, pp.58-87.

¹⁵ Blackburn, R.S., Natural dyes in madder (*Rubia* spp.) and their extraction and analysis in historical textiles. *Coloration Technology*, (2017), 133(6), pp.449-462.

¹⁶ Ahmed, H.E., et al., Identification of natural dyes in rare Coptic textile using HPLC-DAD and mass spectroscopy in museum of Faculty of Arts, Alexandria University, Egypt. *Dyes and Pigments*, (2017), 145, pp.486-492.

¹⁷ Serrano, A., van den Doel, A., van Bommel, M., Hallett, J., Joosten, I. and van den Berg, K.J., 2015. Investigation of crimson-dyed fibres for a new approach on the characterization of cochineal and kermes dyes in historical textiles. *Analytica Chimica Acta*, 897, pp.116-127.

¹⁸ Kramell, A., et al., A fast and reliable detection of indigo in historic and prehistoric textile samples. *Journal of Mass Spectrometry*, (2015), 50(9), pp.1039-1043.

¹⁹ Abdel-Kareem, O.M., Identification of dyestuffs in a rare Coptic garment using high performance liquid chromatography with photodiode array detection (HPLC-PDA). *Journal of Textile and Apparel, Technology and Management*, (2010), 6(3), pp. 1-7.

²⁰ Sanz, E., Arteaga, A., García, M.A., Cámara, C. and Dietz, C., 2012. Chromatographic analysis of indigo from Maya Blue by LC-DAD-QTOF. *Journal of archaeological science*, 39(12), pp.3516-3523.

²¹ Vankar, P.S. and Shukla, D., 2018. Spectrum of colors from reseda luteola and other natural yellow dyes. *J. text. eng. fash. technol*, 4(2), pp.106-119.

²² Mabrouk, N., Archaeometrical study of a rare embroidered and applied tapestry from the Safavid artworks. part I: weaving fibers and dyes. *Mediterranean Archaeology and Archaeometry*, (2020), 20(1), pp.163-171.

²³ Lech, K., Witowska-Jarosz, J. and Jarosz, M., 2009. Saffron yellow: characterization of carotenoids by high performance liquid chromatography with electrospray mass spectrometric detection. *Journal of mass spectrometry*, 44(12), pp.1661-1667.

²⁴ Rosenberg, E., Characterisation of historical organic dyestuffs by liquid chromatography-mass spectrometry. *Analytical and bioanalytical Chemistry*, (2008), 391, pp.33-57.

²⁵ Shahid, M., et al., Analytical methods for determination of anthraquinone dyes, pp.58-87.

heritage textiles is also relevant to the strategies of preventive conservation, since many organic dyes are not stable to light for a long term²⁶.

Recently, chromatographic techniques, such as Ultra/High Performance Liquid Chromatography-Diode Array Detection (U/HPLC-DAD), (U/HPLC-Electrospray Ionization Tandem Mass Spectrometry (U)HPLC-ESI-MS/MS, or HPLC-Atmospheric Pressure Chemical Ionization Tandem Mass Spectrometry (HPLC-APCI-MS/MS) were approved as standard techniques for the identification of historical dyes. All these chromatographic techniques are based on the extraction of the dyes from their fibers^{27,28,29,30}. Mass spectrometric approaches have been widely used as alternative methods in the identification of historical dyes. These techniques enable collecting information about the analyzed dye from small fibers, with no preparation, extraction, hydrolysis, or derivatization³¹.

The static Time-of-Flight Secondary Ion Mass Spectrometry (TOF-SI-MS) has been widely used in identification of variant polymeric, organic, and biological molecules. In the field of heritage conservation, there were limited applications of TOF-SI-MS, especially identification of food residues and ancient dyes^{32,33}. TOF-SI-MS have been used in the identification of dyes in heritage textiles in the mass range m/z 0-1000. Fragment ions and molecular ions from ancient dyes were identified, and elemental ions from the mordants as well^{34,35}. Matrix-assisted Laser Desorption and Ionization (MALDI) is a soft ionization technique, in which ions can be directly produced from the solid material and has been usually coupled with TOF-MS³⁶. MALDI needs micro-samples (smaller than other techniques), thus, it can be denoted as “nondestructive” technique. Proteinaceous binders in heritage paintings were successfully identified using MALDI-TOF MS. Different types of animal glue were not

²⁶ Casadio, F., et al., Identification of organic colorants in fibers, paints, and glazes by surface enhanced Raman spectroscopy. *Accounts of chemical research*, (2010), 43(6), pp.782-791.

²⁷ Troalen, L.G., et al., Historical textile dyeing with *Genista tinctoria* L.: a comprehensive study by UPLC-MS/MS analysis. *Analytical Methods*, (2014), 6(22), pp.8915-8923.

²⁸ Taujenis, L. and Olšauskaitė, V., Identification of main constituents of historical textile dyes by ultra performance liquid chromatography with photodiode array detection. (2012), *Chemija*, 23(3).

²⁹ Petroviciu, I., Albu, F. and Medvedovici, A., LC/MS and LC/MS/MS based protocol for identification of dyes in historic textiles. *Microchemical Journal*, (2010), 95(2), pp.247-254.

³⁰ Ahmed, H.E., et al., Identification of natural dyes in rare Coptic textile, pp.486-492.

³¹ Kramell, A.E., et al., Fast direct detection of natural dyes, pp.12990-12997.

³² Kim, J.P. and Lee, J.J., Natural dyes in Korea, pp. 27-33.

³³ Kempson, I.M., et al., Time-of-flight secondary ion mass spectrometry analysis of hair from archaeological remains. *European Journal of Mass Spectrometry*, (2003), 9(6), pp.589-597.

³⁴ Mabrouk, N., Investigation and analysis of two Coptic textile fragments in the agricultural museum in Egypt. *Shedet*, (2020), 7(7), pp.277-297.

³⁵ Kim, J.P. and Lee, J.J., 2003. Natural dyes in Korea, pp. 25-38.

³⁶ Schiller, J., et al., Matrix-assisted laser desorption and ionization time-of-flight (MALDI-TOF) mass spectrometry in lipid and phospholipid research. *Progress in lipid research*, (2004), 43(5), pp.449-488.

Comparative Study Between MALDI-TOF-MS and HPLC-DAD-TOF-MS Techniques in Identifying Dyes in A Heritage Caucasian Carpet

distinguished, due to the same structure and sequences of the collagens^{37,38}. It has been used in the identification of microbial strains in heritage books³⁹, identification of adhesives “cellulose ethers” used in the conservation of heritage materials⁴⁰, and discrimination of Acacia gums in works of art⁴¹.

2. Materials and Methods

2.1. Case study carpet

The case study carpet is a heritage Caucasian one, dated back to 18th century, preserved in the Heritage Collection of the Agricultural Museum in Egypt (Fig. 1A). It is rectangular in shape, around 1.7*4.25 m size.

The carpet is made of woolen knits, dyed with different colors. It is decorated with different motifs, including dragon, floral and geometric motifs (Fig. 1B). It suffers many deterioration features such as a rupture and tears (Fig. 1C), different stains, dirt, abrasion of the knots (Fig. 1D), deteriorative display technique which caused the distortion of the shape (Fig. 1A). It is in bad need of urgent treatment. 8 microsamples in different colors were collected from the carpet verso for technical investigation and analysis; B1 dark blue, B2 light blue, Y1 brilliant yellow, Y2 pale-yellow, R1 red, O1 orange, Br1 brown, Bc1 black).

³⁷ Romero-Pastor, J., et al., Application of Matrix-Assisted Laser Desorption and Ionization Time of Flight Mass Spectrometry to the Study of the Proteinaceous Binders in Paint: Blue Paint Composition in the Series “The Life of Virgin” by Alonso Cano (17th Century) as a Case Study. *European Journal of Mass Spectrometry*, (2015), 21(1), pp.19-26.

³⁸ Kuckova, S., Hynek, R. and Kodicek, M., 2007. Identification of proteinaceous binders used in artworks by MALDI-TOF mass spectrometry. *Analytical and Bioanalytical Chemistry*, 388, pp.201-206.

³⁹ Kraková, L., et al., Comparison of methods for identification of microbial communities in book collections: Culture-dependent (sequencing and MALDI-TOF MS) and culture-independent (Illumina MiSeq). *International Biodeterioration & Biodegradation*, (2018), 131, pp.51-59.

⁴⁰ Ruiz-Recasens, C., et al., Identification of cellulose ethers in cultural heritage by means of MALDI-TOF-MS. *Journal of Cultural Heritage*, (2017), 24, pp.53-59.

⁴¹ Granzotto, C., et al., Discrimination of Acacia gums by MALDI-TOF MS: applications to micro-samples from works of art. *Microchemical Journal*, (2019), 144, pp.229-241.



Fig. 1 The case study of carpet; A) the whole carpet on a hanger, B) details in the right corner, C) ruptures and tears, D) knots abrasion and staining.

2.2. Chemicals

Methyl alcohol (MeOH) and acetonitrile (CAN) (Merck, (Darmstadt, Germany) were used as mobile phase components of HPLC grade. Hydrofluoric acid (HF) 48% (Sigma-Aldrich, Steinheim, Germany), dimethyl sulfoxide (DMSO) of ACS grade (Merck Darmstadt, Germany). The plant raw materials (madder, weld, indigo, sumac) from Kremer Pigmente (Aichstetten, Germany) were used as reference for the heritage carpet samples. Apigenin, luteolin, luteolin 7-O- β -D-glucoside, and fistein of HPLC purity (Sigma-Aldrich Steinheim, Germany) were used as standards of dyes identification.

2.3. Extraction of dyes

Around 0.3 mg of each dyed fiber was used to extract its dye from fibers in a 500 μ L solution of 0.4 M HF/DMSO/ACN/MeOH (2:1:1:1, v/v/v/v) in an ultrasonic bath at 40°C for 1h. The extract was centrifuged for 5 min. at 9000 rpm, and the supernatants were evaporated under a stream of nitrogen. The residues were then dissolved in a 300 μ L mixture containing DMSO/ACN/MeOH, (1:1:1, v/v/v), 2 μ L of it were injected into the HPLC column. The plant raw materials were extracted according to the first mixture (HF/DMSO/ACN/MeOH (2:1:1:1, v/v/v/v), while the standards were dissolved in the second mixture (DMSO/ACN/MeOH (1:1:1, v/v/v))⁴².

⁴² Otlowska, O., et al., Chromatographic and spectroscopic identification and recognition of natural dyes, uncommon dyestuff components, and mordants: Case study of a 16th century carpet with chintamani motifs. *Molecules*, (2018), 23(2), p.339.

Comparative Study Between MALDI-TOF-MS and HPLC-DAD-TOF-MS Techniques in Identifying Dyes in A Heritage Caucasian Carpet

2.4. Matrix-assisted Laser Desorption/ionization Time-of-Flight Mass Spectrometry (MALDI-TOF-MS)

MALDI-TOF mass spectrometer (Autoflex Speed Bruker, Germany) equipped with 337 nm nitrogen laser in negative reflector mode. The mass spectra for each microfiber were taken between 0-5000 m/z . The collected laser shots for each spectrum were processed with Bruker Flex III, Bruker Xtof software and mMass 5.0.1 software (Bruker)⁴³. Before measurement, the instrument was calibrated by using a commercial peptide mixture MPep. Around 3 mm length microfiber of each dyed thread was directly fixed by adhesive tape to the steel plate of the holder, without any prior preparation or extraction.

2.5. High-Performance Liquid Chromatography-Mass Spectrometry (HPLC-DAD-TOF-MS)

The study used a liquid chromatograph (HPLC) series 1290 (Agilent Technology, Waldbronn, Germany), the diode-array detector (DAD) G1315C, and the triple quadrupole mass spectrometer (QQQ MS) G6470 with an Agilent Jet Stream electrospray ionization source, the autosampler G4226A, with the binary pump G4220A, the thermostated column compartment G1316C. The Agilent MassHunter software (B.07.05, Agilent, Santa Clara, CA, USA) was used to control the system. The extracts' components were separated on a C18 column. The results were monitored with a DAD and a MS connected and characterized with retention times, UV-vis, and mass spectra. The identified dyes were confirmed by HPLC-ESI(-)-QTOF using an Agilent 1290 LC system, coupled with the Agilent QTOF MS G6540 (Santa Clara, CA, USA) in negative ionization mode in the same conditions.

The HPLC separation parameters are as follows: injection vol. 2 mL, column temperature 40 °C, flow rate 0.4 mL min⁻¹, eluents (A) 0.1 % HCOOH in water, (B) 0.1 % HCOOH in ACN/MeOH (1:1; v/v). UV-Vis detection: wavelengths UV: 254, 350 nm; Vis: 580 nm, peak width > 0.1 min (2 s). ESI MS detection: polarity negative, peak width 0.07 min, mode Profile 50-1000 m/z , voltage of fragmentor 100, 200 V, flow rate of drying gas 5 L min⁻¹, temperature of drying gas 300 °C, temperature of heath gas 250 °C, flow rate of sheath gas 11 L min⁻¹, nebulizer pressure 45 psi, capillary voltage 3500 V⁴⁴.

2.6. Scanning Electron Microscopy with Energy Dispersive X-ray spectroscopy (SEM-EDX)

To identify the types of fibers and the probably used mordants, scanning electron microscopy with energy dispersive X-ray spectroscopy (SEM-EDX) (JEOL JSM-6510 LV, JEOL Ltd., Japan) was used, as it is the most common way used in the investigation and analysis of heritage materials⁴⁵. The parameters were as follows:

⁴³ Romero-Pastor, J., et al., Application of Matrix-Assisted Laser Desorption and Ionization Time of Flight Mass Spectrometry, pp.19-26.

⁴⁴ Otlowska, O., et al., Chromatographic and spectroscopic identification, p.339.

⁴⁵ Darchuk, L., et al., *Argentinean prehistoric pigments' study by combined SEM/EDX and molecular spectroscopy*. Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy, (2010). **75**(5): p. 1398-1402.

ACCEL 20KV, MAG 4300, SIGNAL SEI, WD10mm, at low vacuum without coating. SEM-EDX analyses were implemented SEM-EDX.

3. Results and Discussion

3.1. SEM-EDX results

The SEM micrographs (Fig. 2) revealed that all the investigated microfibrs representing different dyed fibers in the case study carpet are made of natural wool fibers as commonly used in heritage textiles and carpets⁴⁶. These fibers are suffering from brittleness and much dust and stains covering the fiber surface. The EDX results revealed the presence of C, N, O, Fe, Na, Mg, Al, Si, S, Cl, K, and Ca elements were in all samples. It clearly reflects how much dust and soiling matter are covering the surface of the fibers. According to the obtained results, alum or ferrous sulphate is probably used as mordant^{47,48}.

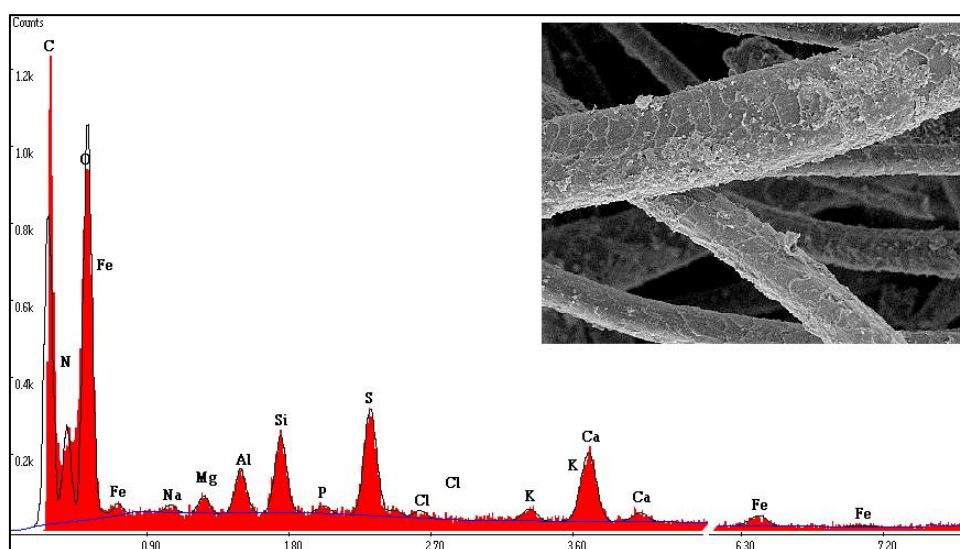


Fig 2. The SEM-EDX results of the microfibrs collected from the case study carpet.

3.2. MALDI-TOF-MS results

The MALDI-TOF-MS obtained results are shown in table (1) and Fig. 3. The spectra of both dark blue samples B1 (Fig.3A) and light blue B2 (Fig.3 B) revealed the presence of indigotin/indirubin mass/charge 262 (m/z 262), indicating using the indigo (*Indigofera tinctoria*) or woad (*Isatis tinctoria*) dye in coloring the fibers in different shadows of blue. Both dyes (indigo and woad) include the same dyeing components, especially indigotin which is the predominant molecule in vat dye. Many other

⁴⁶ Shibayama, N., Wypyski, M. and Gagliardi-Mangilli, E., Analysis of natural dyes and metal threads used in 16 th-18 th century Persian/Safavid and Indian/Mughal velvets by HPLC-PDA and SEM-EDS to investigate the system to differentiate velvets of these two cultures. *Heritage Science*, (2015), 3, pp.1-20.

⁴⁷ Mabrouk, N., Archaeometrical study of a rare embroidered and appliqued tapestry from the Safavid artworks. part I: weaving fibers and dyes. *Mediterranean Archaeology and Archaeometry*, (2020), 20(1), pp.163-171.

⁴⁸ Uring, P., et al., The Bayeux embroidery: a dust deposition assessment. *Heritage Science*, (2018), 6, pp.1-9.

Comparative Study Between MALDI-TOF-MS and HPLC-DAD-TOF-MS Techniques in Identifying Dyes in A Heritage Caucasian Carpet

unknown components appeared at different m/z values. The spectrum of brilliant yellow Y1 sample (Fig.3 C) revealed the presence of apigenin (m/z 270) and luteolin (m/z 286), indicating the use of the Weld (*Reseda luteola*) dye, or probably the use of dyer's broom (*Genista tinctoria*) in coloring the fibers in the brilliant yellow dye⁴⁹. both dyes (weld and dyer's broom) include luteolin and apigenin, only the presence of genistin will define which dye was used⁵⁰. The spectrum of pale-yellow Y2 (Fig.3 D) revealing no identification of any components, only two high intensity peaks were observed, but no capability of identification, hence, no dye was suggested to be used. Probably, no dye was used, and this pale-yellow color is attributed to the use of wool fibers of pale-yellow sheep, or probably due to the natural aging process^{51,52,53}.

The spectra of samples redR1 (Fig.3 E) and orang O1 (Fig.3 F) revealed the presence of Alizarin (m/z 240), and Purpurin (m/z 256) indicating the use of madder (*Rubia tinctorum*) dye in coloring the fibers in both red and orange colors⁵⁴. Different concentration of madder:water ratio, or using different mordants, may cause these variant colors, or the orange color might be originally red, but it has changed to orange due to the natural aging^{55,56}. Regarding the brownBr1 (Fig.3 G) spectrum, it revealed the presence of only Alizarin (m/z 240), indicating the probable use of madder (*Rubia tinctorum*) dye. The black Bc1 (Fig.3 H) spectrum revealed no identification of any components, three low intensity peaks were observed, but no capability of identification, hence, no dye was suggested to be used. We can interpret this result as using natural wool fibers of a black-color sheep. All the unknown components are attributed to some degradation occurred in dyes, or the incapability of the instrument to identify them⁵⁷.

⁴⁹ Selvius DeRoo, C. and Armitage, R.A., Direct identification of dyes in textiles, pp.6924-6928.

⁵⁰ Mantzouris, D., et al., HPLC–DAD–MS analysis of dyes identified in textiles from Mount Athos. *Analytical and bioanalytical chemistry*, (2011), 399(9), pp.3065-3079.

⁵¹ Mabrouk, N., Archaeometrical study of a rare embroidered and appliqued tapestry, pp.163-171.

⁵² Abdel-Kareem, O., et al., Monitoring the environmental conditions and their role in deterioration of textiles collection in museum of faculty of archaeology, Cairo university. *Egypt. Sci. Cult*, (2022), 8, pp.77-88.

⁵³ Cooper, H.R., Yellowing of textiles due to atmospheric pollutants. *Textile progress*, (1987), 15(4), pp.1-6.

⁵⁴ Abdel-Kareem, O.M., Identification of dyestuffs in a rare Coptic garment, pp. 1-7.

⁵⁵ Kramell, A.E., et al., Mapping natural dyes in archeological textiles by imaging mass spectrometry. *Scientific reports*, (2019), 9(1), p.2331.

⁵⁶ Mabrouk, N., Archaeometrical study of a rare embroidered and appliqued tapestry, pp.163-171.

⁵⁷ Romero-Pastor, J., et al., Application of Matrix-Assisted Laser Desorption and Ionization Time of Flight Mass Spectrometry, pp.19-26.

Table (1) MALDI-TOF-MS results of different microfibers of the case study carpet; dark blue (B1), light blue (B2), brilliant yellow (Y1), pale-yellow (Y2), red (R1), orange (O1), brown (Br1), black (Bc1).

No .	Symb ol	Color	a.i.	m/z	Proposed Identification	Molecular Formula	Suggested dye	Biological source
1	B1	Dark blue	55.5	244	Unknown	Unknown	Indigo/ woad	<i>Indigofera tinctoria</i> / <i>Isatis tinctoria</i>
			258.1	262	Indigotin/	C ₁₆ H ₁₀ N ₂ O ₂		
			75	279	Unknown	Unknown		
			58.6	300	Unknown	Unknown		
2	B2	Light blue	20.2	254	Unknown	Unknown	Indigo/ woad	<i>Indigofera tinctoria</i> / <i>Isatis tinctoria</i>
			20.3	255	Unknown	Unknown		
			101.5	262	Indigotin/	C ₁₆ H ₁₀ N ₂ O ₂		
			32.4	293	Unknown	Unknown		
			20.1	300	Unknown	Unknown		
3	Y1	Brilliant yellow	372	270	Apigenin	C ₁₅ H ₁₀ O ₅	Weld/ Dyer's broom	<i>Reseda luteola</i> / <i>Genista tinctoria</i>
			290.3	286	Luteolin	C ₁₅ H ₁₀ O ₆		
4	Y2	Pale-yellow	108.1	276	Unknown	Unknown	Unknown	Unknown
			99.7	309	Unknown	Unknown		
5	R1	Red	243	240	Alizarin	C ₁₄ H ₈ O ₄	Madder	<i>Rubia tinctorum</i>
			235.2	256	Purpurin	C ₁₄ H ₈ O ₅		
			90.7	502	Unknown	Unknown		
6	O1	Orange	243	240	Alizarin	C ₁₄ H ₈ O ₄	Madder	<i>Rubia tinctorum</i>
			235.2	256	Purpurin	C ₁₄ H ₈ O ₅		
7	Br1	Brown	243.4	240	Alizarin	C ₁₄ H ₈ O ₄	Madder	<i>Rubia tinctorum</i>
			148.1	445	Unknown	Unknown		
8	Bc1	Black	60.4	388	Unknown	Unknown	Unknown	Unknown
			52.6	500	Unknown	Unknown		
			62.7	570	Unknown	Unknown		

Comparative Study Between MALDI-TOF-MS and HPLC-DAD-TOF-MS Techniques in Identifying Dyes in A Heritage Caucasian Carpet

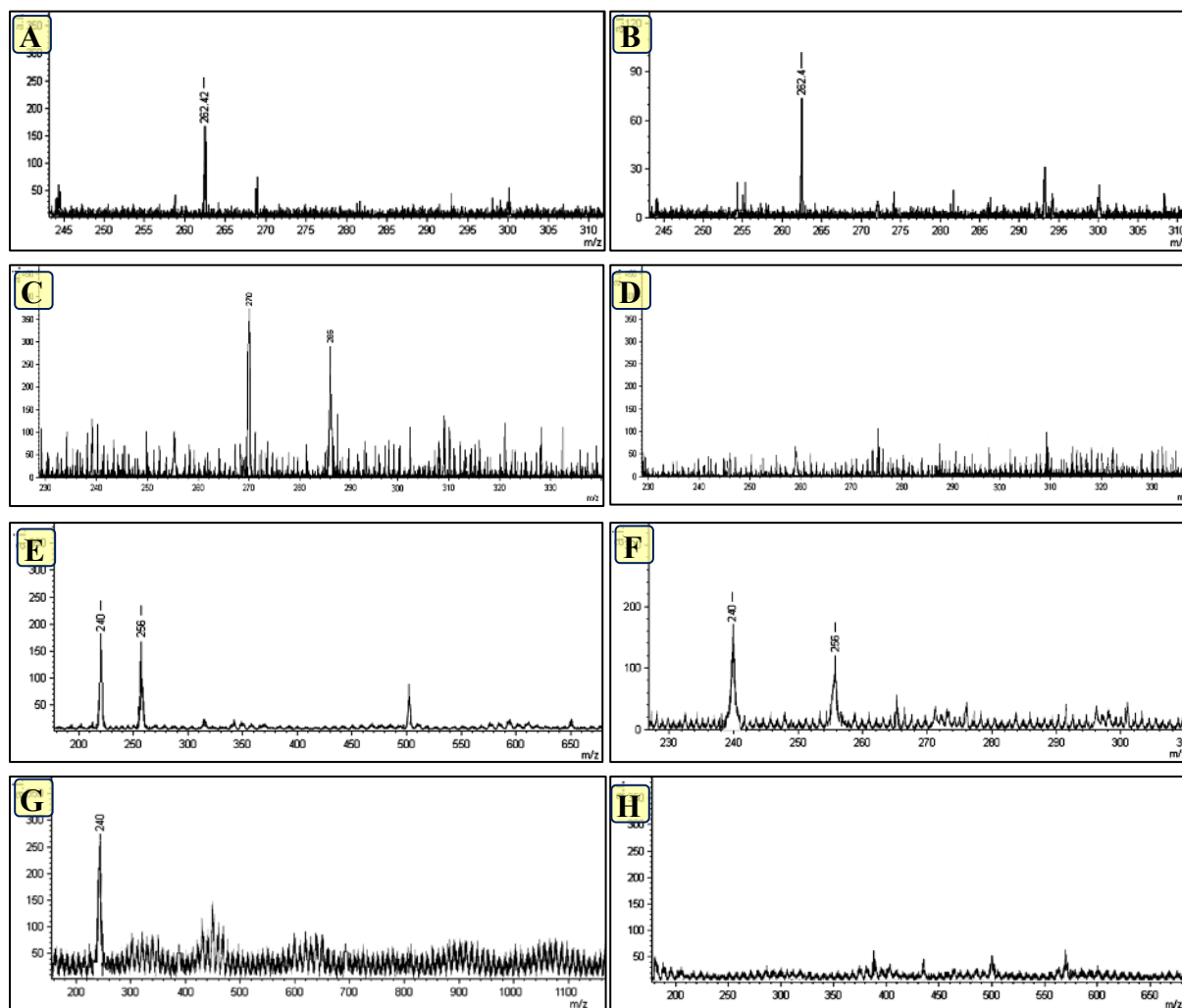


Fig 3. The MALDI-TOF-MS spectra of different microfibers of the case study carpet; dark blue B1 (A), light blue B2 (B), brilliant yellow Y1 (C), pale-yellow Y2 (D), red R1 (E), orange O1 (F), brown Br1(G), black Bc1 (H).

3.3. HPLC-DAD-TOF-MS results

The HPLC-DAD-TOF-MS obtained results are shown in table (2) and Fig. 4. The chromatograms at 350 nm of blue extracts taken from both dark blue sample B1 (Fig.4 A) and light blue sample B2 (Fig.4 B) revealed the presence of Indigotin (m/z 262) at t_R 18.3, and Indirubin (m/z 262) at t_R 19.4. Both indigotin and indirubin reflect the use of the indigo (*Indigofera tinctoria*) or woad (*Isatis tinctoria*) dye in dyeing the blue fibers in different shadows from dark to light^{58,59}. The chromatogram at 350 nm of Brilliant yellow Y1 sample (Fig.4 C) revealed the presence of apigenin-C-diglucoside (m/z 594), luteolin-O-diglucoside (m/z 610), luteolin-3,7'-O-diglucoside (m/z 610), luteolin-7-O-glucoside (m/z 448), apigenin-7-O-glucoside (m/z 423), chryseriol-O-glucoside (m/z 462), luteolin-4'-O-glucoside (m/z 448), luteolin (m/z 286), apigenin (m/z 270), chryseriol (m/z 300). These identified components suggest using weld

⁵⁸ Selvius DeRoo, C. and Armitage, R.A., Direct identification of dyes in textiles by direct analysis in real time-time of flight mass spectrometry. *Analytical chemistry*, (2011), 83(18), pp.6924-6928.

⁵⁹ Kramell, A., et al., A fast and reliable detection of indigo, pp.1039-1043.

(*Reseda luteola*) dye^{60,61}. The chromatogram at 350 nm of pale-yellow (Fig.4 D) revealed the presence of apigenin-C-diglucoside (m/z 594), luteolin-O-diglucoside (m/z 610), luteolin-7-O-glucoside (m/z 448), apigenin-7-O-glucoside (m/z 423), luteolin (m/z 286), apigenin (m/z 270). These identified components suggest using weld (*Reseda luteola*) dye. The absence of genistin compound suggests that the weld (not dyer's broom dye) is the dye used. Both dyes almost include the same compounds, except genistin, in dyer's broom only ⁶².

The spectra of samples red R1 (Fig.4 E), orang O1 (Fig.4 F), and brown Br1 (Fig. 4 G) revealed the common presence of Alizarin (m/z 240), Purpurin (m/z 256), and alizarin O-primeveroside in the three samples, in addition to rare presence of other components, such as munjistin (m/z 284), nordamnacanthal (m/z 268), and hystazarin (m/z 240). These aforementioned components indicate the use of madder (*Rubia tinctorum*) dye in coloring the fibers in red, orange, and brown colors⁶³. The black Bc1 (Fig.4 H) spectrum revealed the presence of ellagic acid (m/z 302) and gallic acid (m/z 170). Both components refer to the probable use of sumac (*Rhus tripartite*) dye, which was commonly used in dyeing heritage textiles in black colors⁶⁴, or may refer to any other tannin-including natural products commonly used (in combination with iron) to achieve brown or black hue in textile dyeing. This is also confirmed by the presence of (Fe) element in the EDX results of the analyzed fibers^{65,66}.

Table (2) HPLC-DAD-TOF-MS spectrochromatographic results of case study heritage dye extracts; dark blue (B1), light blue (B2), brilliant yellow (Y1), pale-yellow (Y2), red (R1), orange (O1), brown (Br1), black (Bc1).

No	Symb ol	Color	tR (min)	m/z	Proposed compound	Molecular Formula	Suggested dye	Biological source
1	B1	Dark blue	18.3	262	Indigotin	C ₁₆ H ₁₀ N ₂ O ₂	Indigo/ woad	<i>Indigofera tinctoria</i> / <i>Isatis tinctoria</i>
			19.4	262	Indirubin	C ₁₆ H ₁₀ N ₂ O ₂		
2	B2	Light blue	18.3	262	Indigotin	C ₁₆ H ₁₀ N ₂ O ₂	Indigo/ woad	<i>Indigofera tinctoria</i> / <i>Isatis tinctoria</i>
			19.4	262	Indirubin	C ₁₆ H ₁₀ N ₂ O ₂		

⁶⁰ Otłowska, O., et al., Chromatographic and spectroscopic identification, p.339.

⁶¹ Peggie, D.A., Hulme, A.N., McNab, H. and Quye, A., Towards the identification of characteristic minor components from textiles dyed with weld (*Reseda luteola* L.) and those dyed with Mexican cochineal (*Dactylopius coccus* Costa). *Microchimica Acta*, (2008), 162, pp.371-380.

⁶² Mantzouris, D., et al., HPLC–DAD–MS analysis of dyes identified in textiles, pp.3065-3079.

⁶³ Kramell, A.E., et al., Mapping natural dyes in archeological textiles, p.2331.

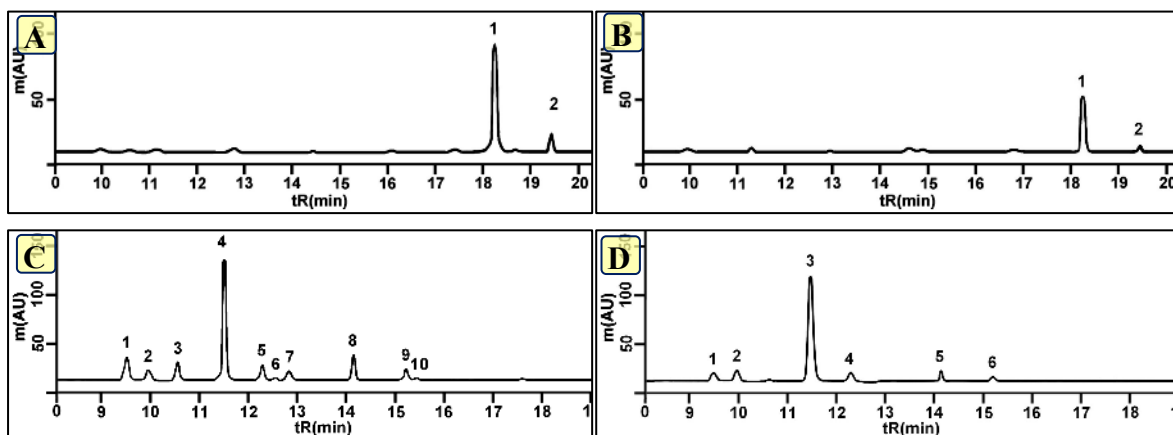
⁶⁴ Baaka, N., Sumac (*Rhus Tripartite*): A natural dye used for simultaneous coloration and functional finishing on textiles. *Journal of Natural Fibers*, (2022), 19(13), pp.7265-7274.

⁶⁵ Manhita, A., et al., Unveiling the colour palette of Arraiolos carpets: Material study of carpets from the 17th to 19th century period by HPLC-DAD-MS and ICP-MS. *Journal of Cultural Heritage*, (2014), 15(3), pp.292-299.

⁶⁶ Mantzouris, D., et al., HPLC–DAD–MS analysis of dyes identified in textiles, pp.3065-3079.

Comparative Study Between MALDI-TOF-MS and HPLC-DAD-TOF-MS Techniques in Identifying Dyes in A Heritage Caucasian Carpet

3	Y1	Brilliant yellow	9.2	594	apigenin-C-	$C_{27}H_{30}O_{15}$	Weld	<i>Reseda luteola</i>
			9.6	610	luteolin-O-diglucoside	$C_{27}H_{30}O_{16}$		
			10.2	610	luteolin-3,7'-O-	$C_{27}H_{30}O_{16}$		
			11.3	448	luteolin-7-O-glucoside	$C_{21}H_{20}O_{11}$		
			12.3	423	apigenin-7-O-	$C_{21}H_{20}O_{10}$		
			12.6	462	chryseriol-O-	$C_{22}H_{22}O_{11}$		
			12.4	448	luteolin-4'-O-glucoside	$C_{21}H_{20}O_{11}$		
			14.1	286	Luteolin	$C_{15}H_{10}O_6$		
			15.6	270	apigenin	$C_{15}H_{10}O_5$		
			15.8	300	chryseriol	$C_{16}H_{12}O_6$		
4	Y2	Pale-yellow	9.2	594	apigenin-C-	$C_{27}H_{30}O_{15}$	Weld	<i>Reseda luteola</i>
			9.6	610	luteolin-O-diglucoside	$C_{27}H_{30}O_{16}$		
			11.3	448	luteolin-7-O-glucoside	$C_{21}H_{20}O_{11}$		
			12.3	423	apigenin-7-O-	$C_{21}H_{20}O_{10}$		
			14.1	286	Luteolin	$C_{15}H_{10}O_6$		
			15.6	270	apigenin	$C_{15}H_{10}O_5$		
5	R1	Red	13	534	alizarin O-	$C_{25}H_{26}O_{13}$	Madder	<i>Rubia tinctorum</i>
			15.1	240	hystazarin	$C_{14}H_8O_4$		
			14.7	284	munjistin	$C_{15}H_8O_6$		
			16.4	240	alizarin	$C_{14}H_8O_4$		
			18.2	256	purpurin	$C_{14}H_8O_5$		
6	O1	Orange	13	534	alizarin O-	$C_{25}H_{26}O_{13}$	Madder	<i>Rubia tinctorum</i>
			16.4	240	alizarin	$C_{14}H_8O_4$		
			18.2	256	purpurin	$C_{14}H_8O_5$		
			20.6	268	nordamnacanthal	$C_{15}H_8O_5$		
7	Br1	Brown	13	534	alizarin O-	$C_{25}H_{26}O_{13}$	Madder	<i>Rubia tinctorum</i>
			15.1	240	hystazarin	$C_{14}H_8O_4$		
			14.7	284	munjistin	$C_{15}H_8O_6$		
			16.4	240	alizarin	$C_{14}H_8O_4$		
			18.2	256	purpurin	$C_{14}H_8O_5$		
			20.6	268	nordamnacanthal	$C_{15}H_8O_5$		
8	Bc1	Black	11.5	302	ellagic acid	$C_{14}H_6O_8$	Sumac	<i>Rhus tripartita</i>
			2.7	170	gallic acid	$C_7H_6O_5$		



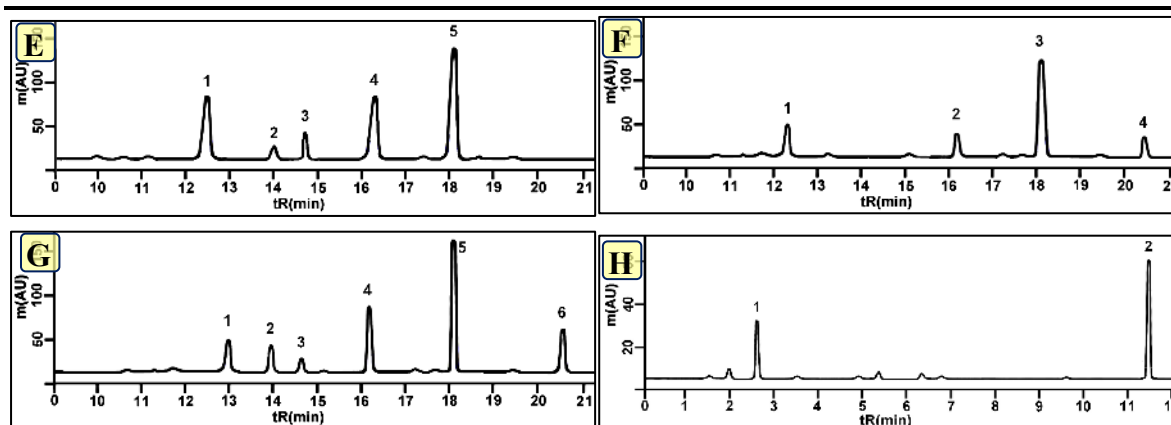


Fig. 4. HPLC-DAD-TOF-MS chromatograms at 350 nm of case study heritage dye extracts; dark blue (A), light blue (B), brilliant yellow (C), pale-yellow (D), red (E), orange (F), brown (G), black (H).

The comparison between the results of both instruments is sometimes in favor of the one or the other. MALDI-TOF-MS has proved many advantages^{67,68}: 1) Easy use and operation without any sample extraction and/or preparation. 2) No sampling, it is considered as a non-invasive or micro-invasive technique, it needs only around 3 mm length of one fiber, which is too small to fly in air or fall on ground. 3) No chemicals required, so it is safe and less hazardous. 4) Cost-effective and faster, due to no sample extraction and/or preparation, no required chemicals.

The only disadvantage is the less accuracy due to less identified compounds. For example, in the brilliant yellow (Y1), it revealed the presence of apigenin and luteolin, both compounds are the main ingredients in both Weld (*Reseda luteola*) dye and dyer's broom (*Genista tinctoria*)⁶⁹, that's why we couldn't chemically confirm which dye was used, especially we couldn't confirm the absence of genistin, the distinct compound of dyer's broom⁷⁰. HPLC-DAD-TOF-MS not only confirmed the absence of genistin, but also identified extra eight compounds in the dye. Regarding the pale-yellow Y2 sample, MALDI-TOF-MS revealed no capability of identification of any components, consequently, we attributed the pale-yellow color to the use of natural pale-yellow sheep wool fibers, but this wasn't right. HPLC-DAD-TOF-MS revealed six compounds of weld ingredients, thus it was confirmed that it isn't a natural pale-yellow sheep wool, but weld dye was used to color in pale-yellow shadows.

Regarding the red, orange, and brown dyes, MALDI-TOF-MS identified only alizarin and sometimes purpurin. Despite the absence of other extra compounds identified by HPLC-DAD-TOF-MS, the MALDI-TOF-MS is appeared suitable to identify natural madder, because no other red dye materials contain alizarin and purpurin.

⁶⁷ Kraková, L., et al., Comparison of methods for identification of microbial communities, pp.51-59.

⁶⁸ Granzotto, C., et al., Discrimination of Acacia gums, pp.229-241.

⁶⁹ Selvius DeRoo, C. and Armitage, R.A., Direct identification of dyes in textiles by direct analysis in real time-time of flight mass spectrometry. *Analytical chemistry*, (2011), 83(18), pp.6924-6928.

⁷⁰ Mantzouris, D., et al., HPLC-DAD-MS analysis of dyes identified in textiles, pp.3065-3079.

Comparative Study Between MALDI-TOF-MS and HPLC-DAD-TOF-MS Techniques in Identifying Dyes in A Heritage Caucasian Carpet

purpurin^{71,72} that's why, MALDI-TOF-MS is initially recommended in case of only need to identify the dye, not to characterize the material or study all components and degradation products. As for the black sample Bc1, MALDI-TOF-MS revealed no identified components, thus, we also attributed the black color to the use of natural wool of a black-color sheep, but this wasn't right. HPLC-DAD-TOF-MS revealed the presense of ellagic acid and gallic acid, thus it was confirmed that it isn't a natural wool of black-color sheep, but sumac dye was suggested⁷³.

4. Conclusion

The results obtained from MALDI-TOF-MS compared to HPLC-DAD-TOF-MS revealed many final remarks: 1) MALDI-TOF-MS is easy to use and without any sample extraction and/or preparation, it is a non-invasive or micro-invasive technique which need only micro-fiber, no chemicals required, so it is safe and less hazard, cost-effective and faster, due to no sample extraction and/or preparation, no required chemicals, etc. 2) the only disadvantage of MALDI-TOF-MS is the less accuracy, it can identify some (not all) components in the analyzed fibers. This shortage was clear and repeated, resulting sometimes in no identification of many dyes, due to no compounds being absolutely identified, and sometimes caused misunderstanding which dyeing material was used in case of identified compounds. 3) On the contrary, HPLC-DAD-TOF-MS is very accurate, and identify all/most sample components, resulting in defined identification of the dyes used. 4) It has some disadvantages as well, e.g. it is difficult to use, needs bigger sample, requires many extractions and preparation process, considered a destructive or micro-destructive technique, high cost, and time consuming, due to the sample preparation and the required chemicals.

MALDI-TOF-MS is recommended in the case of only need to identify the dye, not to characterize the whole material and different related degradation products, it can help in a direct way especially in the dyes that were repeatedly identified like madder. It is also recommended in the identification of the blue dye, which is either indigo or woad, and it proves capability of identification. It is also recommended as the first step of dyes identification due to its easiness, fastness, cost-effectiveness, micro-fibers required, etc., hereafter, and in case of failure to identify, HPLC-DAD-TOF-MS is recommended as the second step, or in case of more details are required. HPLC-DAD-TOF-MS is recommended in distinction between yellow flavonoids dyes, especially weld, dyer's broom, and fustic.

5. Acknowledgments

The author would like to thank Dr. S. Kuckova, Charles University in Czech Republic for the MALDI-TOF-MS. Thanks to Dr. M. Sliwka-Kaszyńska at Gdansk University of Technology in Poland for the HPLC-DAD-TOF-MS results. Thanks to

⁷¹ Ford, L., et al., Mild extraction methods using aqueous glucose solution for the analysis of natural dyes in textile artefacts dyed with Dyer's madder (*Rubia tinctorum* L.). *Journal of Chromatography A*, (2017), 1487, pp.36-46.

⁷² Blackburn, R.S., Natural dyes in madder, pp.449-462.

⁷³ Baaka, N., Sumac (*Rhus Tripartita*), pp.7265-7274.

Prof. H. Elmitwalli for HPLC-DAD-QqQ-MS lab at our Faculty of Archaeology Damietta University.

6. References

- 1) Abdel-Kareem, O., Samaha, S., El-Nagar, K., Essa, D. and Nasr, H., Monitoring the environmental conditions and their role in deterioration of textiles collection in museum of faculty of archaeology, Cairo university. *Egypt. Sci. Cult*, (2022), 8, pp.77-88.
- 2) Abdel-Kareem, O.M., Identification of dyestuffs in a rare Coptic garment using high performance liquid chromatography with photodiode array detection (HPLC-PDA). *Journal of Textile and Apparel, Technology and Management*, (2010), 6(3), pp. 1-7.
- 3) Ahmed, H.E., Tahoun, I.F., Elkholy, I., Shehata, A.B. and Ziddan, Y., Identification of natural dyes in rare Coptic textile using HPLC-DAD and mass spectroscopy in museum of Faculty of Arts, Alexandria University, Egypt. *Dyes and Pigments*, (2017), 145, pp.486-492
- 4) Baaka, N., Sumac (*Rhus Tripartita*): A natural dye used for simultaneous coloration and functional finishing on textiles. *Journal of Natural Fibers*, (2022), 19(13), pp.7265-7274.
- 5) Blackburn, R.S., Natural dyes in madder (*Rubia* spp.) and their extraction and analysis in historical textiles. *Coloration Technology*, (2017), 133(6), pp.449-462.
- 6) Casadio, F., Leona, M., Lombardi, J.R. and Van Duyne, R., Identification of organic colorants in fibers, paints, and glazes by surface enhanced Raman spectroscopy. *Accounts of chemical research*, (2010), 43(6), pp.782-791.
- 7) Cooper, H.R., Yellowing of textiles due to atmospheric pollutants. *Textile progress*, (1987), 15(4), pp.1-6.
- 8) Darchuk, L., Tsybrii, Z., Worobiec, A., Vázquez, C., Palacios, O. M., Stefaniak, *Argentinean prehistoric pigments' study by combined SEM/EDX and molecular spectroscopy*. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, (2010). **75**(5): p. 1398-1402.
- 9) De Ferri, L. and Andriulo, F., Non-Invasive Analytical Techniques for the Study of Dyes and Pigments in Historical Textiles. *Handbook of Museum Textiles*, 2, (2022), pp.171-190.
- 10) Ford, L., Henderson, R.L., Rayner, C.M. and Blackburn, R.S., Mild extraction methods using aqueous glucose solution for the analysis of natural dyes in textile artefacts dyed with Dyer's madder (*Rubia tinctorum* L.). *Journal of Chromatography A*, (2017), 1487, pp.36-46.
- 11) Gillard, R.D., Hardman, S.M., Thomas, R.G. and Watkinson, D.E., The detection of dyes by FTIR microscopy. *Studies in conservation*, (1994), 39(3), pp.187-192.
- 12) Granzotto, C., Sutherland, K., Arslanoglu, J. and Ferguson, G.A., Discrimination of Acacia gums by MALDI-TOF MS: applications to micro-samples from works of art. *Microchemical Journal*, (2019), 144, pp.229-241.
- 13) Karapanagiotis, I., Abdel-Kareem, O., Kamaterou, P. and Mantzouris, D., Identification of dyes in coptic textiles from the museum of faculty of archaeology, Cairo university. *Heritage*, (2021), 4(4), pp.3147-3156.
- 14) Kempson, I.M., Skinner, W.M., Kirkbride, P.K., Nelson, A.J. and Martin, R.R., Time-of-flight secondary ion mass spectrometry analysis of hair from archaeological remains. *European Journal of Mass Spectrometry*, (2003), 9(6), pp.589-597.
- 15) Kim, J.P. and Lee, J.J., 2003. Natural dyes in Korea. Seoul: Seoul National University Press. Korea (2003).
- 16) Kraková, L., Šoltys, K., Otlewska, A., Pietrzak, K., Purkrťová, S., Savická, D., Puškárová, A., Bučková, M., Szemes, T., Budiš, J. and Demnerová, K., Comparison of methods for

Comparative Study Between MALDI-TOF-MS and HPLC-DAD-TOF-MS Techniques in Identifying Dyes in A Heritage Caucasian Carpet

- identification of microbial communities in book collections: Culture-dependent (sequencing and MALDI-TOF MS) and culture-independent (Illumina MiSeq). *International Biodeterioration & Biodegradation*, (2018), 131, pp.51-59.
- 17) Kramell, A., Porbeck, F., Kluge, R., Wiesner, A. and Csuk, R., A fast and reliable detection of indigo in historic and prehistoric textile samples. *Journal of Mass Spectrometry*, (2015), 50(9), pp.1039-1043.
 - 18) Kramell, A.E., Brachmann, A.O., Kluge, R., Piel, J. and Csuk, R., Fast direct detection of natural dyes in historic and prehistoric textiles by flowprobe™-ESI-HRMS. *RSC Advances*, (2017), 7(21), pp.12990-12997.
 - 19) Kramell, A.E., García-Altares, M., Pötsch, M., Kluge, R., Rother, A., Hause, G., Hertweck, C. and Csuk, R., Mapping natural dyes in archeological textiles by imaging mass spectrometry. *Scientific reports*, (2019), 9(1), p.2331.
 - 20) Kuckova, S., Hynek, R. and Kodicek, M., 2007. Identification of proteinaceous binders used in artworks by MALDI-TOF mass spectrometry. *Analytical and Bioanalytical Chemistry*, 388, pp.201-206.
 - 21) Lech, K., Witowska-Jarosz, J. and Jarosz, M., 2009. Saffron yellow: characterization of carotenoids by high performance liquid chromatography with electrospray mass spectrometric detection. *Journal of mass spectrometry*, 44(12), pp.1661-1667
 - 22) Mabrouk, N., Archaeometrical study of a rare embroidered and applique leather tapestry from the Safavid artworks. part I: weaving fibers and dyes. *Mediterranean Archaeology and Archaeometry*, (2020), 20(1), pp.163-171.
 - 23) Mabrouk, N., Investigation and analysis of two Coptic textile fragments in the agricultural museum in Egypt. *Shedet*, (2020), 7(7), pp.277-297
 - 24) Manhita, A., Balcaen, L., Vanhaecke, F., Ferreira, T., Candeias, A. and Dias, C.B., Unveiling the colour palette of Arraiolos carpets: Material study of carpets from the 17th to 19th century period by HPLC-DAD-MS and ICP-MS. *Journal of Cultural Heritage*, (2014), 15(3), pp.292-299.
 - 25) Mantzouris, D., Karapanagiotis, I., Valianou, L. and Panayiotou, C., HPLC–DAD–MS analysis of dyes identified in textiles from Mount Athos. *Analytical and bioanalytical chemistry*, (2011), 399(9), pp.3065-3079.
 - 26) Nedumpillil, N.N., Sankaran, A., Jose, S., George, S. and Thomas, S., Destructive analytical techniques for the analysis of historic textiles. *Handbook of Museum Textiles*, (2022), 2, pp.215-234.
 - 27) Otlowska, O., Ślebioda, M., Kot-Wasik, A., Karczewski, J. and Śliwka-Kaszyńska, M., Chromatographic and spectroscopic identification and recognition of natural dyes, uncommon dyestuff components, and mordants: Case study of a 16th century carpet with chintamani motifs. *Molecules*, (2018), 23(2), p.339.
 - 28) Peggie, D.A., Hulme, A.N., McNab, H. and Quye, A., Towards the identification of characteristic minor components from textiles dyed with weld (*Reseda luteola* L.) and those dyed with Mexican cochineal (*Dactylopius coccus* Costa). *Microchimica Acta*, (2008), 162, pp.371-380.
 - 29) Petroviciu, I., Albu, F. and Medvedovici, A., LC/MS and LC/MS/MS based protocol for identification of dyes in historic textiles. *Microchemical Journal*, (2010), 95(2), pp.247-254.
 - 30) Poulin, J., A new methodology for the characterisation of natural dyes on museum objects using gas chromatography–mass spectrometry. *Studies in Conservation*, (2018), 63(1), pp.36-61.
 - 31) Pour, R.A., Gashti, M.P. and He, J., State-of-the-Art Characterization Methods for Historic Textiles. *Handbook of Museum Textiles*, (2022) 2, pp.107-129.

- 32) Romero-Pastor, J., Navas, N., Rodriguez-Simon, L., Lario-Simón, A., Kuckova, S. and Manzano, E., Application of Matrix-Assisted Laser Desorption and Ionization Time of Flight Mass Spectrometry to the Study of the Proteinaceous Binders in Paint: Blue Paint Composition in the Series “The Life of Virgin” by Alonso Cano (17th Century) as a Case Study. *European Journal of Mass Spectrometry*, (2015), 21(1), pp.19-26.
- 33) Rosenberg, E., Characterisation of historical organic dyestuffs by liquid chromatography–mass spectrometry. *Analytical and bioanalytical Chemistry*, (2008), 391, pp.33-57.
- 34) Ruiz-Recasens, C., Campo-Frances, G., Fernandez-Vidal, I. and Oriola, M., Identification of cellulose ethers in cultural heritage by means of MALDI-TOF-MS. *Journal of Cultural Heritage*, (2017), 24, pp.53-59.
- 35) Schiller, J., Süß, R., Arnhold, J., Fuchs, B., Lessig, J., Müller, M., Petković, M., Spalteholz, H., Zschörnig, O. and Arnold, K., Matrix-assisted laser desorption and ionization time-of-flight (MALDI-TOF) mass spectrometry in lipid and phospholipid research. *Progress in lipid research*, (2004), 43(5), pp.449-488.
- 36) Selvius DeRoo, C. and Armitage, R.A., Direct identification of dyes in textiles by direct analysis in real time-time of flight mass spectrometry. *Analytical chemistry*, (2011), 83(18), pp.6924-6928.
- 37) Serrano, A., van den Doel, A., van Bommel, M., Hallett, J., Joosten, I. and van den Berg, K.J., 2015. Investigation of crimson-dyed fibres for a new approach on the characterization of cochineal and kermes dyes in historical textiles. *Analytica Chimica Acta*, 897, pp.116-127
- 38) Shahid, M., Wertz, J., Degano, I., Aceto, M., Khan, M.I. and Quye, A., Analytical methods for determination of anthraquinone dyes in historical textiles: A review. *Analytica Chimica Acta*, (2019), 1083, pp.58-87.
- 39) Shibayama, N., Wypyski, M. and Gagliardi-Mangilli, E., Analysis of natural dyes and metal threads used in 16 th-18 th century Persian/Safavid and Indian/Mughal velvets by HPLC-PDA and SEM-EDS to investigate the system to differentiate velvets of these two cultures. *Heritage Science*, (2015), 3, pp.1-20.
- 40) Sanz, E., Arteaga, A., García, M.A., Cámara, C. and Dietz, C., 2012. Chromatographic analysis of indigo from Maya Blue by LC–DAD–QTOF. *Journal of archaeological science*, 39(12), pp.3516-3523
- 41) Taujenis, L. and Olšauskaitė, V., Identification of main constituents of historical textile dyes by ultra performance liquid chromatography with photodiode array detection. (2012), *Chemija*, 23(3).
- 42) Tiedemann, E.J. and Yang, Y., Fiber-safe extraction of red mordant dyes from hair fibers. *Journal of the American Institute for Conservation*, (1995), 34(3), pp.195-206.
- 43) Troalen, L.G., Phillips, A.S., Pegg, D.A., Barran, P.E. and Hulme, A.N., Historical textile dyeing with *Genista tinctoria* L.: a comprehensive study by UPLC-MS/MS analysis. *Analytical Methods*, (2014), 6(22), pp.8915-8923.
- 44) Uring, P., Chabas, A., De Reyer, D., Gentaz, L., Triquet, S., Mirande-Bret, C. and Alfaro, S., The Bayeux embroidery: a dust deposition assessment. *Heritage Science*, (2018), 6, pp.1-9.
- 45) Vankar, P.S. and Shukla, D., 2018. Spectrum of colors from reseda luteola and other natural yellow dyes. *J. text. eng. fash. technol*, 4(2), pp.106-119